TECHNICAL BOOKLET

MCLAREN TROPHY AMERICA 2026

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INTRODUCTION - MCLAREN TROPHY AMERICA

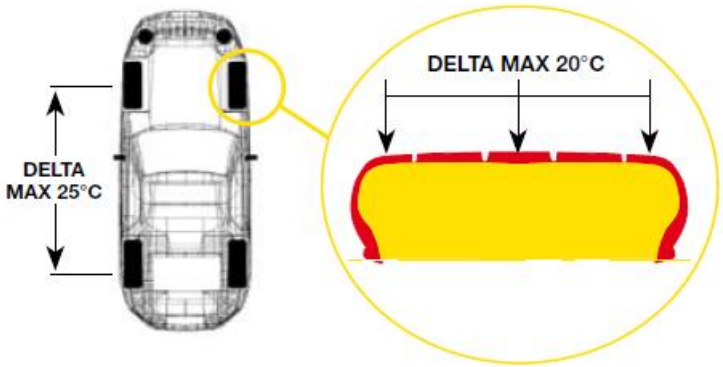
INTRODUCTION
This document provides information regarding the fitment, handling and usage of the tyres on cars competing in this Championship. For detailed technical information please refer to the individual Tyre Data Books for each tyre size, which are included as separate PDF files within this Pirelli Data Package.

MCLAREN TROPHY AMERICA - FITMENT LIST

Make/Model	Front axle				Rear axle		
	Wheel size	Dry tyre	Wet tyre		Wheel size	Dry tyre	Wet tyre
McLaren Artura Trophy EVO	18" x 11J	280/650-18 **DHG**	280/650-18 **WHB**		18" x 12J	300/690-18 **DHG**	300/690-18 **WHB**

GENERAL INFORMATION

OPERATING INSTRUCTIONS	
Before each run	
Pressures	- The tyre pressure must always be over the declared **MINIMUM INFLATION PRESSURE** defined within this book. - Cold tyre pressures must always be set in order that the hot target pressure stated within this book is reached. - Dry air is recommended to inflate tyres to avoid sudden changes in pressure due to humidity.

After each run	
Temperatures	- the offset in measured garage bulk temperature between the inside and outside of the tyre should not exceed 20°C for optimum tyre performance (see Figure 1 below)
	- the offset in measured garage bulk temperatures between front and rear axle should not exceed 25°C for optimum tyre performance (see Figure 1 below)
Pressures	- the measured garage pressure should be not less than the hot target value stated within this book
	- **all pressure limitations (minimum and hot) stated in this book will be monitored during each event, and modified if deemed necessary.**
Figure 1	

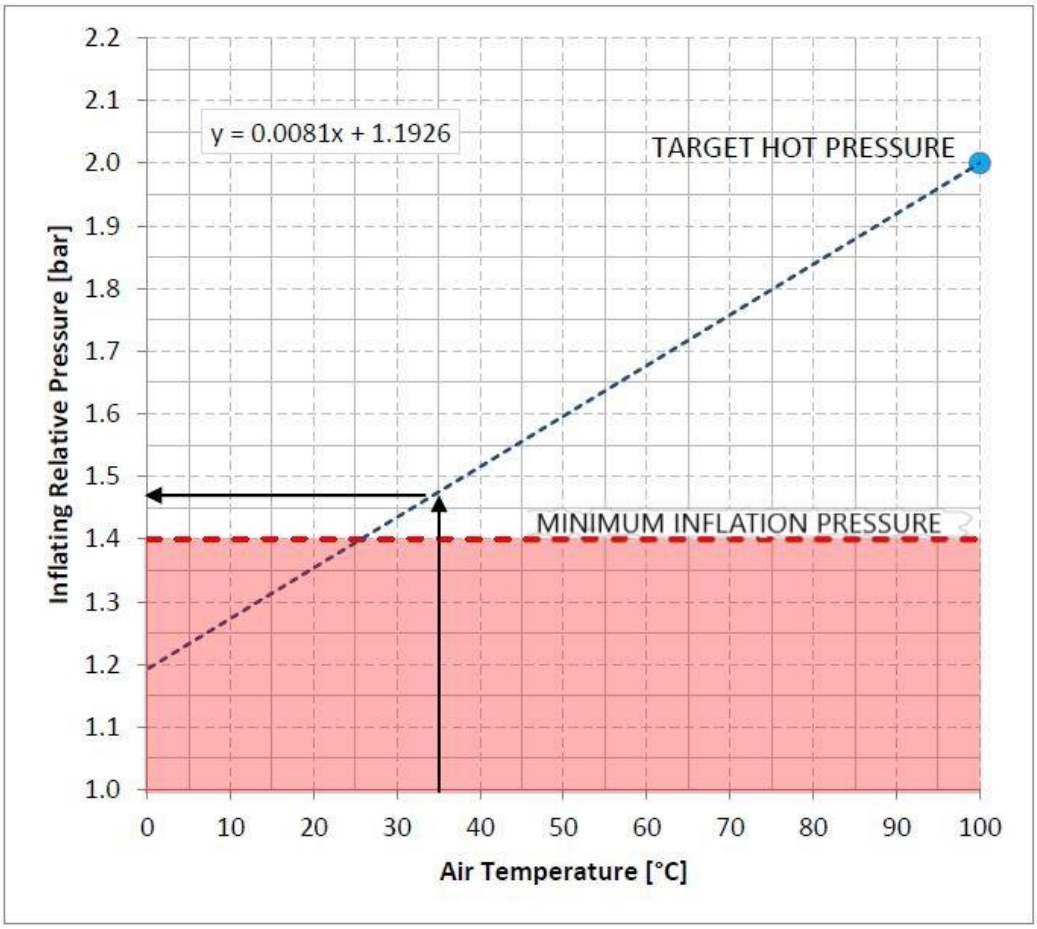
TYRE PRESSURE MANAGEMENT

INFLATION PRESSURE

Tyre pressure increases with temperature (while preheating and when car is running) following the theoretical line shown in *Figure 2,* below.

To inflate tyres in warm-to-hot ambient conditions (air temperature above 25°C), the dashed line in *Figure 2* shows the correct pressure value.

Figure 2:



If the ambient temperature is below 25°C the minimum cold pressure limit is encountered.

The car should not be released onto the circuit if the pressure is below the Minimum Inflation Pressure.

TYRE PRESSURE MANAGEMENT

INFLATION PRESSURE ADJUSTMENTS

In cold conditions, after one lap or as soon as the tyres have warmed-up, a pressure adjustment may be required in order to prevent exceeding target hot pressure.

The pressure adjustment for warmed tyres can be calculated as follows:

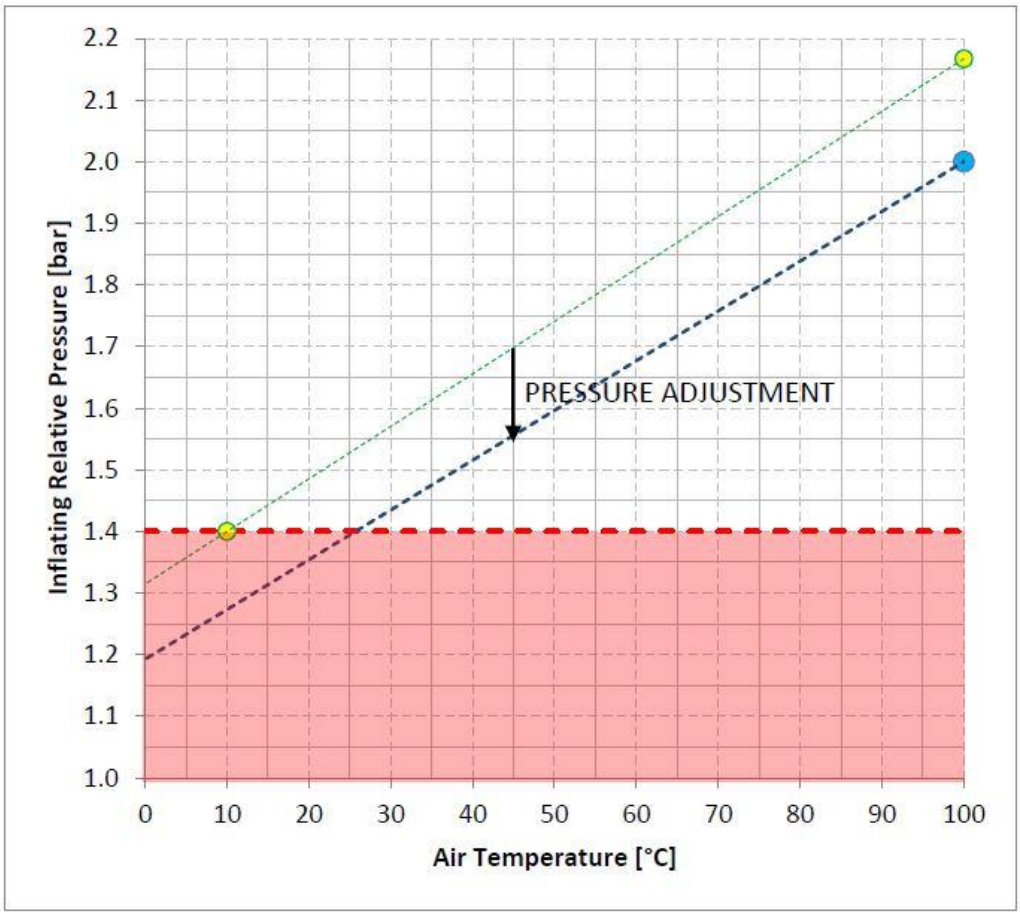
$$T_m[°C] = \frac{(P_m[bar] + 1{,}013) \cdot (T_i[°C] + 273{,}15)}{2{,}413} - 273{,}15$$

$$\Delta P\,[bar] = P_m[bar] - \frac{3{,}013}{373{,}15} \cdot (T_m[°C] + 273{,}15) + 1{,}013$$

Ti [°C]:	inflating air ambient temperature
Pm [bar]:	measured pressure (after warm-up)
Tm [°C]:	calculated air temperature (after warm-up)
ΔP [bar]:	calculated adjustment for pressure

- Always note the ambient temperature when inflating cold tyres to the Minimum Inflation Pressure
- Measure the tyre pressures after warm-up

Figure 3:



RUNNING PARAMETERS - MCLAREN TROPHY AMERICA

TYRE INFLATION PRESSURE			
MINIMUM INFLATION PRESSURE	***1.3 bar***	***/***	***18.9 Psi***
HOT PRESSURE TARGET	*2.0 bar*	*/*	*29 Psi*

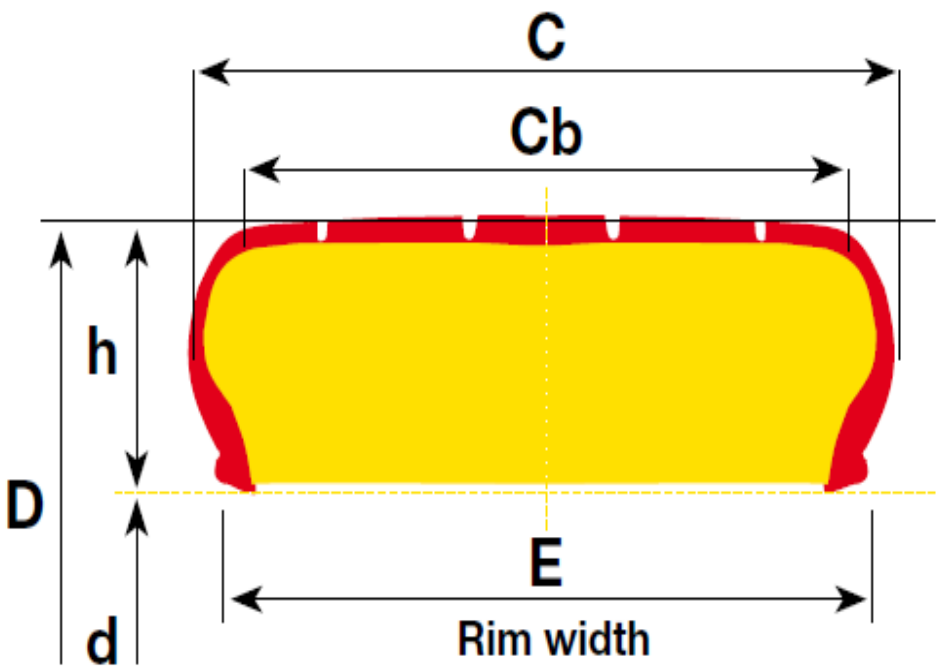
CAR SETUP PARAMETERS	
Maximum Static Negative Camber -Front Axle	-4.0 deg
Maximum Static Negative Camber -Rear Axle	-3.5 deg

Please note: Teams should always ensure that they refer to the Championship's regulations and any technical bulletins regarding tyre pressures and static negative camber limits, if applicable.

NOTES
• Please be aware that not complying with the above running parameters will expose the product to excessive levels of mechanical stress that could affect the integrity of the product.
• Particular care will be needed for any tyres used at previous events, especially those that have already been used outside of our parameters.
• Driving style during the out lap is of utmost importance to minimise the stress generated over the carcass of the tyres. We recommend that drivers always avoid aggressive kerb usage, particularly until the tyres have reached an internal pressure close to their optimum running pressure.
• We are confident you will understand the above requirements and, if necessary, will adjust your procedures accordingly.
• These recommendations are valid unless the tyres have been damaged.
• To avoid any damage to the tyres caused by air leaks, we recommend the following: - use metal valve bodies instead of rubber ones; - check the fixing of the valves and their seal frequently; - check the valve core; - use metal valve caps with integral rubber o-ring seal.

STATIC MEASUREMENT DEFINITIONS

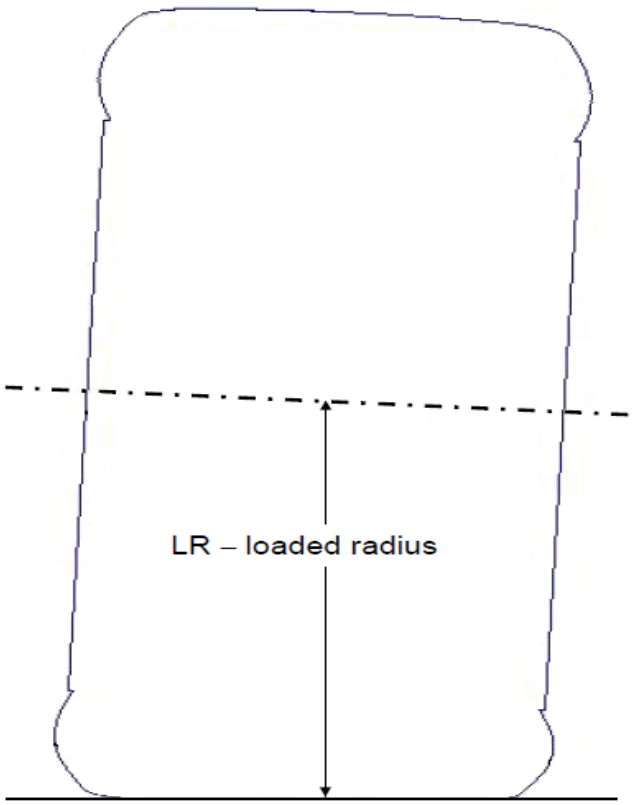
STATIC MEASUREMENTS	
The static measurements within this book are provided for each combination of tyre-rim size. Geometric measurements are taken with the tyre fitted on a rim, inflated to the standard Pirelli running pressure. Please see the individual Tyre Data Books for the pressure per tyre.	
Circumference **πD** [mm]	*length along the middle tread line of the tyre;*
Max width **C** [mm]	*maximum width of the tyre;*
Tread width **Cb** [mm]	*width of the tread.*



A full characterisation of deflection vs. vertical load at different pressures is given; measurements are taken at two different camber levels (0° and -3.0°) for slick tyres.

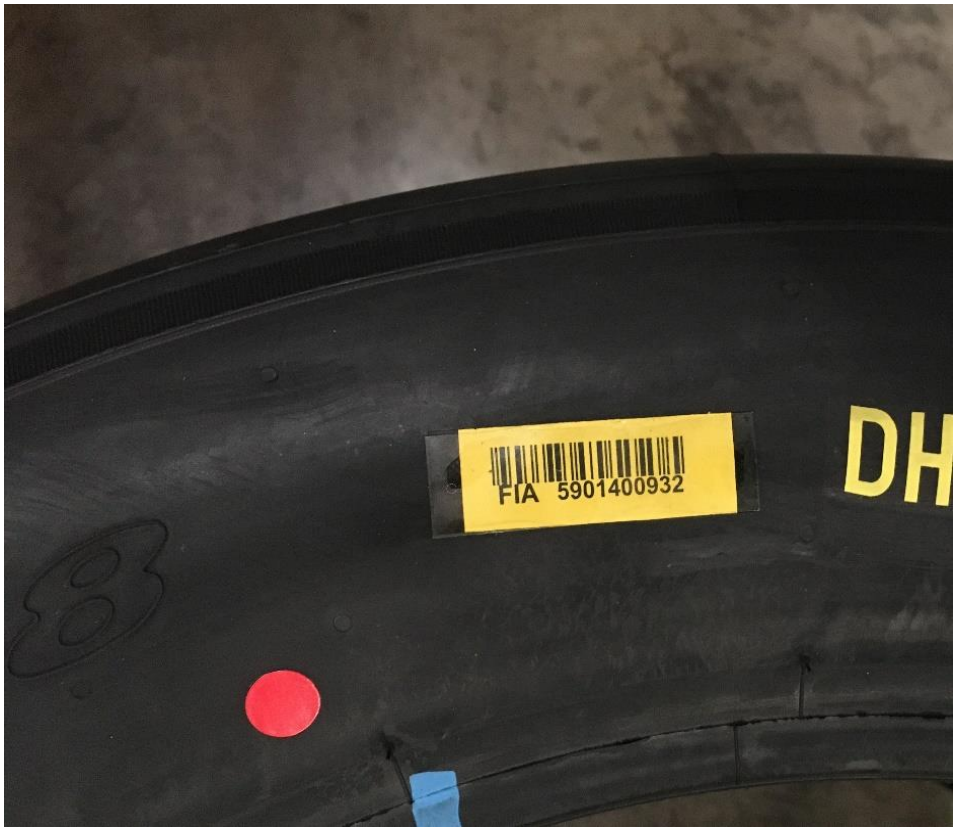
DYNAMIC MEASUREMENT DEFINITIONS

DYNAMIC MEASUREMENTS	
Tyre dynamic characterisation describes changes in dimensions due to speed and vertical load. All measurements are made without any applied camber (0°). Please see the individual Tyre Data Books for the pressure per tyre.	
Loaded radius **L.R.** [mm]	*distance between the wheel center and the ground;*
Rolling radius **R.R.** [mm]	*the length travelled by the tyre for each wheel's revolution divided by 2 π.*



TYRE FITTING - MCLAREN TROPHY AMERICA

FITTING INSTRUCTIONS FOR DHG SPECIFICATIONS

For **DHG** specification products, tyres must be fitted in accordance with the sidewall markings on each tyre. The yellow FIA barcode should be positioned on the outside and the red FIA barcode on the inside. Pirelli personnel will only fit tyres in this prescribed way. Any failure to adhere to this may change the characteristics of the product.

Please note that the red dot on the sidewall should be positioned next to the valve when fitting, for balance reasons.